

BNPL: Smoothing or Debt Trap, and for Whom?

A Research Proposal on the Welfare Effects of Buy Now, Pay Later

Consumer Credit in the United States

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Abstract

Buy Now, Pay Later (BNPL) service is one of the growing payment option worldwide and used by roughly 14 percent of U.S. adults. Some studies find that BNPL helps households spread spending across pay cycles, while others find that adopting it leads to overdraft fees result to debt accumulation. This proposal argues that the disagreement comes from data windows that are too short, household differences, and different products pooled into one category. The research sets out a long-panel event-study design in which each household's effect depends on the liquidity position it held before adoption. The purpose is to identify for whom BNPL smooths consumption and for whom it deepens financial distress. That evidence is relevant to individual households and regulatory policy.

Keywords: Buy Now Pay Later, consumption smoothing, households, credit, event study

1. Introduction

Buy Now, Pay Later (BNPL) has become one of the fastest growing payment options worldwide. The service lets a consumer to split a purchase into few small installments paid over a short period. Global BNPL transaction value in e-commerce reached about \$300 billion in 2025 and is projected to approach \$500 billion by 2030 (Statista, 2026). In the United States, the Consumer Financial Protection Bureau (2022) reports that the number of BNPL loans from five largest lenders rose from 16.8 million in 2019 to 180 million in 2021, while their total value grew from \$2 billion to \$24.2 billion. Over the same 3 years period, Larrimore et al. (2024) show that the share of U.S. adults using BNPL climbed from 10 percent in 2021 to 14 percent in 2023.

Rapid growth of the service is driven by providers such as Klarna, Afterpay, and Affirm, which market BNPL as a simple cash flow tool. Unlike a credit card, it can look more transparent and cheaper because payments on time carry no compounding interest, no revolving balance, and no late fee. However, CFPB (2022) takes a more cautious view to BNPL as part of the consumer credit market and pointing to risks such as overextension, repayment friction, and design features that urge people to spend more. The academic literature is split along the same lines, with studies reaching opposite conclusions about whether BNPL improves welfare or adds to financial strain.

One one hand, Di Maggio et al. (2022) find that liquidity-constrained households use BNPL to smooth spending across pay cycles. By letting a buyer delay payment without drawing on revolving credit, BNPL adds short term flexibility when income and bills do not line up. On the other hand, deHaan et al. (2024) use banking-transaction data for 10.6 million U.S. consumers find that new BNPL users see sharp increases in overdraft charges and credit card fees in the months after they start. These contrasting findings suggest that BNPL may help some households and hurt others.

This household distinction of BNPL become more important to examine broader welfare of BNPL service. Larrimore et al. (2024) report that adults who describe themselves as just getting by or hard to get by are almost twice as likely to use BNPL as those who feel financially comfortable. Among adults who could not cover a \$500 emergency from savings, about one in five used BNPL in 2023. Therefore, a single population average estimate could misstate the product's real effect. If BNPL works differently for fragile households than for stable ones, an average is not enough to judge welfare or to design policy.

This research asks whether the welfare effects of BNPL depend on a household's starting financial position. The central claim is that the disagreement in the literature may simply reflect differences across users. A household with little cash on hand but substantial illiquid wealth may use BNPL to ride out a temporary squeeze. A household with little wealth of any kind may use it in a way that turns into ongoing repayment stress. Both can be labeled liquidity constrained, yet the welfare consequences are very different.

The proposed analysis uses a long-panel event-study design that follows households before and after their first BNPL adoption. The design interacts adoption with each household's baseline liquidity using the typology of Kaplan, Violante, and Weidner (2014). That typology separates the poor hand-to-mouth who hold little wealth of any kind and wealthy hand-to-mouth who hold sizeable illiquid assets such as housing or retirement accounts but little liquid savings. The framework builds on recent BNPL event-study and difference-in-differences work (Di Maggio et al., 2022; deHaan et al., 2024) and applies the heterogeneity-robust estimators of Callaway and Sant'Anna (2021) and de Chaisemartin and d'Haultfœuille (2020).

This design aims to distinguish three mechanisms apart. Under consumption smoothing, distress should dip briefly around adoption and then return to its earlier level. Under overborrowing, distress should climb steadily and stay high. If both happen at the same time, the average can hide the real effect on wealthy hand-to-mouth households and poor hand-to-

mouth households. The proposal therefore studies BNPL's welfare effects by asking for whom the effect is positive and for whom it is negative. This question is important because the rules around BNPL such as CFPB regulation, credit-bureau reporting, and how FICO scores it are all changing quickly. Through this framework, understanding who is helped and who is harmed is also a precondition for helping consumers make better-informed decisions about whether BNPL is a useful tool for their own circumstances.

2. Literature Review

2.1 Theoretical Foundations

Permanent Income Hypothesis (Friedman, 1957) is a theory that forward-looking households try to keep consumption steady even when income arrives unevenly. On this view, more access to short-term credit raises welfare because it lets a household keep spending smooth through a temporary income dip. If consumers behaved exactly as this model assumes, BNPL would be a straightforward improvement.

This theory changes once liquidity constraints are taken seriously. A household is liquidity constrained when it cannot borrow as much as it wants or cannot borrow cheaply at the moment it needs to. Using national survey data on U.S. household finances, Jappelli (1990) found that many households are turned down for credit and others never apply because they expect to be refused. The households blocked from borrowing are usually the ones with low income, little wealth, or no job. These are the households BNPL reaches, since traditional channels are least willing to serve them (Hayashi & Routh, 2025)

Kaplan et al. (2014) also identify that a large share of U.S. households are hand-to-mouth which hold very little liquid wealth even when they own some portion of illiquid assets such as a home or a retirement account. They identify two subtypes that respond to credit in opposite ways. The poor hand-to-mouth hold no wealth of any kind and often use new credit

to replace income they will not get back. The wealthy hand-to-mouth hold sizeable illiquid assets but little cash and use new credit to bridge short cash flow gaps.

Behavioral finance points the other way. Laibson (1997) formalizes hyperbolic discounting about present-biased people put too much weight on today and over-borrow against their future selves. A payment due now feels far heavier than the same amount spread over the coming weeks. Complementing this temporal bias, Thaler (1985) develops the theory of mental accounting that individuals do not evaluate expenditures against their total budget, but instead isolate transactions within narrow mental frames. For example, partitioning a \$100 expense into four distinct \$25 payments feels significantly cheaper because each smaller transaction easily fits within low-impact budget category, even the present value is almost identical. Maesen and Ang (2025) confirm this experimentally that installment framing makes consumers feel less financially constrained and therefore spend more, which is evidence for mental accounting effect. The mechanisms predict that BNPL as product built to make purchases feel more affordable will raise total spending rather than just move its timing.

These three frameworks yield three predictions of BNPL effect. Under the Permanent Income Hypothesis, BNPL access raises welfare. Under a liquidity-constraints model with different starting levels of wealth, BNPL raises welfare for the wealthy hand-to-mouth and is ambiguous for the poor hand-to-mouth. Under the behavioral models, BNPL lifts spending beyond what smoothing alone would justify and may lower welfare even for households that can technically afford the payments.

2.2 The Smoothing Camp

The case for BNPL as a welfare-improving tool rely on Di Maggio, Katz, and Williams (2022) that use banking-transaction data and find liquidity-constrained households use BNPL to even out spending across pay cycles. Papich (2022) adds supporting causal evidence using credit-bureau data and a two-way fixed-effects design that exploits where and when BNPL

became available at a large U.S. retailer. Papich finds that access to BNPL helps lower financial distress that is total past-due amounts fall, current delinquencies drop, and credit scores rise with the largest gains among consumers who began with fair credit scores. This indicates that BNPL helps constrained consumers smooth their spending rather than pile up unsustainable debt.

Recent work from the JPMorgan Chase Institute (Wheat, Henry-Nickie, Kim, Neal, and Francisco, 2026) reaches a similar but more detailed conclusion. Using anonymized banking data linked to credit-bureau records for millions of U.S. homeowners, the authors describe BNPL as a liquidity valve. Use rises exactly when households face cash-flow pressure, such as after a job loss or around a home purchase which fits a smoothing role for households that are temporarily stretched. The study supports the smoothing story, but adds one warning. BNPL use rises most when money is tightest, which leaves less cash for big payments like a mortgage.

A similar study comes from outside the BNPL literature. Levine, Beltramo, Blalock, Cotterman, and Simons (2018) study a randomized experiment in Uganda that offered fuel-efficient cookstoves on installments to some buyers and at full price to others. The setting similar to BNPL framework that is a durable good, a credit-constrained population, and an installment offer. They find that the installment option sharply raised adoption compared with full upfront pricing, with no sign of regret or hardship. The mechanism proves that credit access removed a binding constraint for households that genuinely wanted the product.

2.3 The Overborrowing Camp

The case for BNPL as a debt-accumulation device is anchored by deHaan, Kim, Lourie, and Zhu (2024), who use banking-transaction data for 10.6 million U.S. consumers and find that new users see sharp post-adoption increases in overdraft charges and in credit-card interest and fees. The authors pair an instrumental-variables strategy built on consumers' pre-BNPL shopping habits with a difference-in-differences design. The estimated effects are statistically

significant and economically meaningful. Adopters do not just swap one payment method for another, they take on fees that would not have appeared without BNPL.

Maesen and Ang (2025) find a similar pattern. Using customer-level transaction data from a large U.S. retailer before and after it introduced BNPL, they apply a difference-in-differences design and find adoption raises both how often customers buy and how much they spend per trip, with effects that grow over time. The effects are strongest among smaller-basket customer and among those who relied more heavily on credit cards before adopting, exactly the group for whom this mechanism should be sharpest. Kumar, Salo, and Bezawada (2024) report the same pattern in a separate retailer panel with BNPL adoption lifting both purchase frequency and order values. Neither paper measures debt directly, but their results are what the overborrowing story predicts at the spending stage by making each purchase feel smaller, BNPL leads people to spend more rather than simply re-time spending they had already planned.

How BNPL interacts with other credit sharpens the concern. Guttman-Kenney, Firth, and Gathergood (2023) use UK credit-card records to show that many consumers pay their BNPL installments with a credit card. This turns free short-term installment of BNPL into revolving balances that often carry interest near 20 percent, and the behavior is most common among younger consumers and those in the underprivileged areas. Paying off BNPL with a credit card is a clear sign of overborrowing that is the buyer likely could not afford the purchase in the first place

Furthermore, the CFPB (2022) notes that BNPL differs from traditional credit in ways that raise the risk of overextension. One example is loan stacking, where users take on multiple BNPL debts at a large scale. Drawing on 2022 borrower data, the CFPB (2025) finds that about 63 percent of BNPL users held more than one loan at the same time during the year and roughly 33 percent borrowed from more than one provider. Survey evidence from the Behavioural

Insights Team (2023) in the United Kingdom also finds that around 40 percent of BNPL users have borrowed money to repay existing BNPL as direct sign of the debt-spiral dynamic that overborrowing can create.

2.4 Synthesis

The clash between the two camps reveals how complicated BNPL's effect on households really is. Di Maggio, Katz, and Williams (2022), Papich (2022), and deHaan, Kim, Lourie, and Zhu (2024) each point in different directions. The disagreement comes down to three things about how the studies were designed.

First, the time horizons are too short. Every published empirical paper on BNPL relies on one to two years of data because the product is new and long individual-level panels do not yet exist outside proprietary banking datasets. The mechanism itself run on different timeframe. Smoothing is short-run by design and shows up as lower spending volatility in the first few pay cycles. Overborrowing builds slowly as households roll forward unpaid balances, miss payments, and set off overdraft chains. A short study weighted toward early adopters can catch the smoothing benefit and miss the distress that only surfaces after two or three years. Studies that watch different windows therefore tend to find different things and none observes the full path.

Second, differences across users go unmodeled. The Kaplan et al. (2014) typology implies that wealthy hand-to-mouth households are where BNPL should produce the cleanest smoothing benefit, while poor hand-to-mouth households are where an overborrowing dynamic should bite hardest. A study that estimates one average effect across both groups blends effects of opposite sign, so its result depends on the mix of households in the sample rather than on any real truth about the product. Few studies split households by their starting cash position, and those that do find sharp differences between groups (Fig. 1)

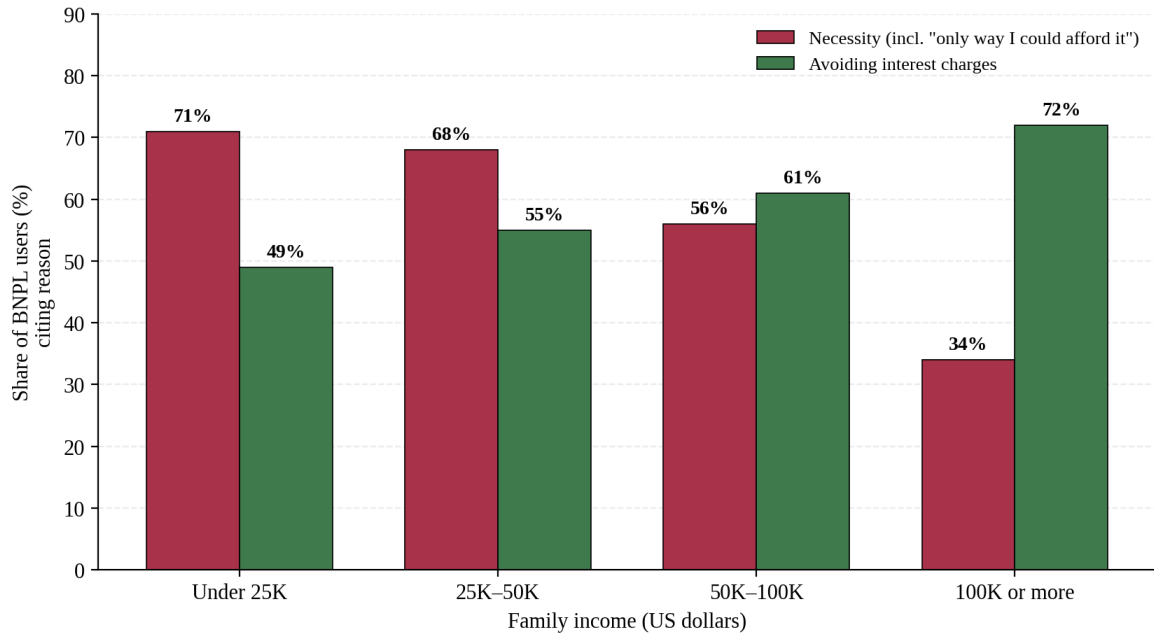


Fig 1. Reasons for BNPL use, by family income. Source: Larrimore et al. (2024), FEDS Notes, Table 7. Author's reconstruction.

Third, differences across products are collapsed. Klarna, Afterpay, Affirm, and Apple Pay Later are not the same product. They differ in fees, in whether late payments reach credit bureaus, in underwriting intensity, in default penalties, and in how installment plans are structured. Studies that pool all BNPL transactions into one bucket erase this variation.

Table 1. Major Empirical BNPL Studies and the Source of Disagreement

Study	Data	Method	Horizon	Main finding
Di Maggio, Katz & Williams (2022)	Banking transactions; BNPL adopters	DiD / IV on rollout	Short-run	Smooths spending for constrained users, but also raises total spending.
Papich (2022)	Credit-bureau data; retailer rollout	Two-way fixed effects	Short-run	Access lowers distress and slightly raises credit scores.
deHaan, Kim, Lourie & Zhu (2024)	10.6M U.S. consumers; bank data	DiD with IV	Short to medium	Raises overdraft and credit-card fees after adoption.
Maesen & Ang (2025)	Retailer transactions + experiments	DiD + experiments	Retail window	Installments raise how often and how much people buy.

Study	Data	Method	Horizon	Main finding
Wheat et al. (2026)	Bank + credit-bureau; homeowners	Descriptive panel	2019–2025	BNPL acts as a liquidity valve; use peaks when budgets are tight.
Larrimore et al. (2024)	Fed SHED + credit merge	Descriptive / panel	2021–2023	Constrained consumers are more likely to use BNPL out of need.
CFPB market report (2025)	6 large pay-in-four providers	Aggregate transaction data	2019–2023	Late-fee and charge-off rates fell in 2023 versus 2022.

Note: The table summarizes the logic of the literature rather than claiming that the studies estimate identical quantities.

3. Data and Method

3.1 Why a Standard Difference-in-Differences is Not Enough

A plain difference-in-differences (DiD) design would compare the change in outcomes for adopters, before and after adoption, with the change for non-adopters over the same window. Under the parallel trends assumption, the difference between these two changes identifies the average treatment effect on the treated. For BNPL, this design fails for two reasons, each of which motivates a specific feature of the design proposed below.

First, selection bias is severe. Adopters are not a random subset of consumers. Larrimore et al. (2024) show that adults who started using BNPL in 2023 were already more likely to report financial stress in the prior year, so a simple before-and-after comparison would blame BNPL for a decline that was already underway and overstate the harm. The pre-existing trend is an unobserved confounder that a two-period DiD cannot detect. An event study fixes this by showing the years before adoption explicitly, so the researcher can check whether treated and control households were already drifting apart.

Second, a single pooled effect is uninformative once heterogeneity is taken seriously. Even a correctly identified average can hide different effects for wealthy hand-to-mouth and

poor hand-to-mouth households. The average blends two opposite findings, which is less helpful for the specific question of whom BNPL affects and over what time horizon.

3.2 Proposed Design: Event Study with Heterogeneity

The proposed design is a dynamic event study, extended with an explicit interaction by baseline liquid-asset position. The baseline specification is:

$$Y_{it} = \alpha_i + \lambda_t + \sum (k \neq -1) \beta_k D_{i,t,k} + X_{it}' \gamma + \varepsilon_{it}$$

Here Y_{it} is the outcome for household i in year t , such as overdraft fees, credit-card balance, savings depletion, delinquency, or self-reported financial stress. The household fixed effect α_i is a per-household baseline that absorbs everything about a household that does not change over time, including permanent income, personality, and general financial habits, so the estimate reflects only what changes within a household, such as adopting BNPL. The year fixed effect λ_t absorbs economy-wide shocks that hit everyone, such as inflation or pandemic-era transfers. Writing t_{i*} for the year household i first adopts, $k = t - t_{i*}$ is the number of years since adoption. The indicator $D_{i,t,k}$ equals one when household i is exactly k years from adoption and zero otherwise; for non-adopters it is always zero.

The coefficients β_k are the dynamic treatment effects of interest. A positive β_k means that, k years after adoption, the distress outcome rose relative to the year before adoption, which is evidence of overborrowing; a negative value means distress fell, which is evidence of smoothing; a value near zero means no change. The coefficients are normalized to $\beta_{-1} = 0$, which sets the year just before adoption as the starting line, so every other β_k reads as a deviation from that point. That pre-adoption year is a clean anchor because the household has not yet been treated. The vector X_{it} holds time-varying controls such as age, income, household size, and employment status, and standard errors are clustered at the household level.

The heterogeneity extension adds an interaction with baseline constraint status:

$$Y_{it} = \alpha_i + \lambda_t + \sum \beta_k D_{i,t,k} + \sum \delta_k (D_{i,t,k} \times \text{Constrained}_i) + X_{it}' \gamma + \varepsilon_{it}$$

Constrained_i indicates hand-to-mouth status at baseline, defined following Kaplan, Violante, and Weidner (2014) as holding liquid assets below half of monthly income that split further into poor and wealthy hand-to-mouth. The measured at baseline and held fixed, so adopting BNPL cannot move a household between groups. The δ_k coefficients are the core of the contribution because they measure whether constrained households follow a different post-adoption path than unconstrained adopters. Table 2 makes this concrete by showing the predicted effect one year after adoption, at event time $k = +1$ for four household types.

Table 2. Predicted Effect One Year After Adoption, by Household Type

Household type	D (k = +1)	Constrained	Predicted treatment effect
Constrained adopter	1	1	$\beta_{+1} + \delta_{+1}$
Unconstrained adopter	1	0	β_{+1}
Constrained non-adopter	0	1	0
Unconstrained non-adopter	0	0	0

A positive δ_k on overdraft fees, for example would mean that constrained households fare worse than unconstrained ones at the same point after adoption. That is the direct empirical version of the heterogeneity prediction the literature has not yet tested.

3.3 Hypotheses and Predicted Patterns

Three theoretical frameworks produce three hypotheses with different prediction for the pattern of estimated coefficients.

Table 3. Hypotheses and Predicted Event-Study Patterns

Hypothesis	Predicted pattern	Interpretation
H1: Pure consumption smoothing	$\beta_k < 0$ for $k = 0, 1$; back to 0 by $k = 3$	BNPL eases short-term cash pressure without lasting debt.

Hypothesis	Predicted pattern	Interpretation
H2: Pure overborrowing	$\beta_k > 0$ and rising across k	Distress builds after adoption and persists.
H3: Heterogeneous effects	$\delta_k > 0$ at all post-adoption horizons	Constrained households fare worse than unconstrained adopters (BNPL smooths for some and traps others)

Note: distress outcomes include overdraft charges, credit-card fees, delinquencies, savings depletion, and self-reported financial stress.

Hypothesis H₃ is the central claim that the proposal expect to find. The literature already contains evidence consistent with H₁ (Di Maggio et al., 2022; Papich, 2022) and with H₂ (deHaan et al., 2024; Maesen and Ang, 2025), but no study has tested whether the two opposite findings come from looking at different kinds of households. The proposed design tests this directly. Finding H₃ would settle the disagreement by showing each camp is right for a different group. Finding the opposite that no difference between groups would move the debate forward by ruling out the most likely explanation. Fig. 2 sketches what each hypothesis would look like.

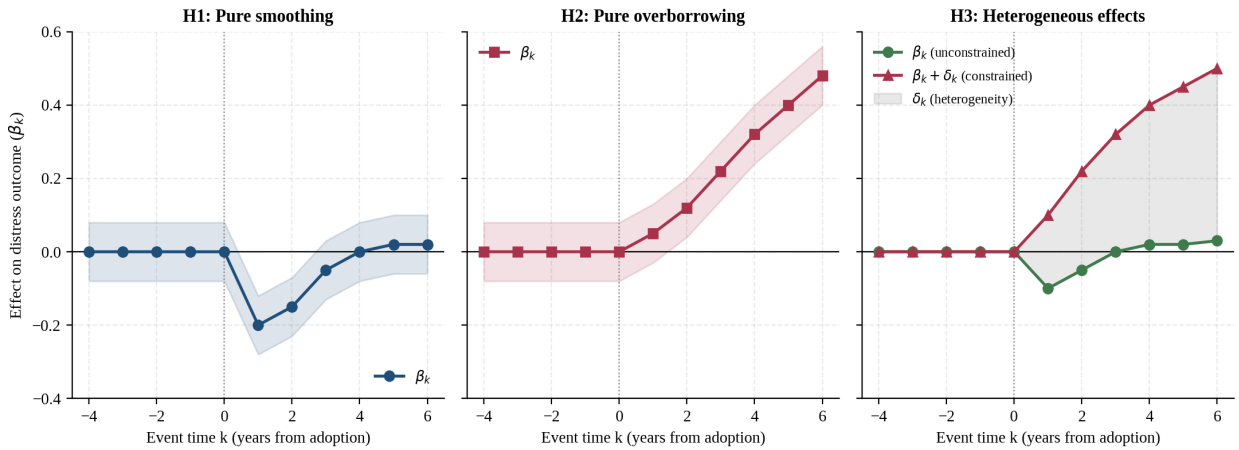


Fig. 2. Schematic event-study paths under the three hypotheses (figure is illustrative, not estimated)

3.4 Data Sources and Threats to Identification

The proposed data combine the Panel Study of Income Dynamics (PSID) with the Federal Reserve's Survey of Household Economics and Decisionmaking (SHED). The PSID provides long panel needed to measure effects beyond two years includes detailed balance sheet and detailed questions on household assets, debts, and financial stress. The SHED supplies the BNPL adoption indicator about BNPL use and frequency since 2021 (Larrimore et al., 2024).

The two surveys are not formally linked, which is the practical challenge and the two ways to bridge them differ in how confidently they assign BNPL use. Imputation gives each PSID household a probability of using BNPL, estimated from traits observed in both surveys such as age, income, and financial fragility. Another method by synthetic matching to finds the single most similar SHED respondent for each PSID household and copies that respondent's actual adoption status, it is more definite but adds error when the match is imperfect. The design would run both and treat agreement between them as a robustness check.

The ideal data would be the proprietary banking-transaction panels used by deHaan et al. (2024) and Papich (2022), which offer direct view of BNPL transactions. These are the right benchmark for future work but are not public, so the proposed design is built around what a researcher without proprietary access can credibly do.

Four threats to identification deserve attention. The first is selection on unobservables that adopters may differ from non-adopters in ways that change over time and the household fixed effect cannot absorb, such as a recent income shock that hits in the same year as adoption. The standard check is to inspect the pre-adoption coefficients (β_{-4} , β_{-3} , β_{-2}) for flat trends. If they are flat, treated and control households were on the same path before adoption, which supports a causal inference. If they are not flat, the design must be read as descriptive rather than causal, meaning it can show that outcomes changed after adoption but cannot claim BNPL caused the change.

The second is heterogeneous adoption timing. As BNPL spread quickly between 2020 and 2023, households that adopted in different years faced different economic conditions. This can be eased by restricting the sample to a narrow adoption cohort, at the cost of some statistical power. The third threat comes from the statistical method itself. The standard difference-in-differences approach (two-way fixed effects) can give biased results when households adopt at different times, because it ends up using already treated households as a comparison group for later adopters. Recent work from Goodman-Bacon (2021) shows the standard estimate hides many smaller comparisons, some of which count in the wrong direction. Sun and Abraham (2021) show it can create fake pre-trends in an event study and Baker, Larcker, and Wang (2022) show how often it has changed published results. The design therefore uses newer estimators built for staggered adoption. The event-study coefficients are estimated with the Sun and Abraham (2021) estimator, and the headline effect with the Callaway and Sant'Anna (2021) estimator, which compares each group only to households that have not yet adopted.

The fourth threat is rapid policy and reporting change. In 2024 the CFPB issued an interpretive rule treating some BNPL accounts as credit cards under Regulation Z, then signaled in 2025 that it would not enforce it (CFPB, 2025). Over the same period FICO moved to fold BNPL into credit scores and Affirm began reporting loans to Experian (Hayashi & Routh, 2025). These shifts can change consumer incentives and the visibility of BNPL debt, any design spanning these dates should include policy-period indicators or robustness checks around the reporting changes.

With these caveats, the design identifies the path of adopters' financial outcomes under stated assumptions with explicit attention to differences by baseline liquidity. It does not identify the unconditional causal effect of BNPL access on a general population, and it does not replace the proprietary-data work the literature ultimately needs.

Taking these limitations into account, the design can trace how adopters' financial outcomes change over time, with separate results by starting cash position. It cannot prove BNPL's overall effect on the average consumer and it does not replace the richer studies using private bank data that the field still needs.

4. Expected Contribution and Policy Implications

4.1 Conceptual and Empirical Contribution

Conceptually, the proposal shifts the question from whether BNPL helps or harms people on average to who it helps and who it harms. Empirically, it shows how dynamic event-time coefficients can separate a temporary effect from long-term distress. This separation helps cross-sectional finding that BNPL users are financially vulnerable does not show that BNPL caused the vulnerability and a short-run rise in spending does not by itself show a gain in welfare.

The proposal also clarifies what welfare means for different households. For a constrained household that still holds substantial assets, BNPL can raise welfare if it allows smoothing without credit-card interest or payday-loan fees. For a present-biased or severely constrained household, the same product can lower welfare if small installments hide the combined weight of several loans.

4.2 Policy Implications

Some outcome that can help the regulatory follow from the existing evidence even before this study proposed. Each targets failure of the product as currently designed and may help improve the service in the future, especially by addressing the welfare effects of BNPL for different groups.

First, BNPL providers should report to credit bureaus on a consistent basis, which would remove the spiral debt problem documented by the CFPB (2022). Inconsistent reporting keeps both lenders and consumers from seeing total BNPL exposure and that information gap

is enough to misallocate credit no matter which camp is right. Recent industry moves toward folding BNPL into scores and bureau reporting (Hayashi & Routh, 2025) suggest the sector is drifting this way, but consistent treatment across all providers remains a regulatory rather than a voluntary outcome.

Second, loan stacking across providers should be capped. The CFPB (2025) finds that many BNPL users hold simultaneous plans across providers, each small but jointly significant, with about 63 percent holding more than one loan at once and roughly 33 percent borrowing from more than one provider. No single provider can see the total and overdraft consequences documented by deHaan et al. (2024) which concentrate among the households most likely to be vulnerable. The finding that around 40 percent of users have borrowed to repay existing BNPL debt (Behavioural Insights Team, 2023) suggests that some stacking reflects compounding necessity rather than informed choice.

Third, minimum underwriting standards should be required. The absence of meaningful underwriting at checkout is what makes BNPL so widely accessible to financially vulnerable consumers and it also increase the risk of financial stress for significant share of this users. A minimum standard does not require settling the welfare debate, it requires only acknowledging that some adopters are made worse off in ways providers could foresee.

A stronger stance should still stop short of an outright ban, which would strip a potentially useful liquidity tool from consumers who have no access to cheaper credit. The better goal is to preserve the smoothing function while limiting the behavioral and informational ways the product can fail. Standardized repayment disclosures, cross-provider reporting of active loans, and minimum underwriting would target the debt-trap channel without removing the product for households that use it responsibly.

4.3 Limitations

The design would identify the dynamic effects of BNPL adoption on financial stress outcomes, under a parallel-trends assumption and allowing for differences by baseline liquidity. It cannot prove BNPL's overall effect on the average consumer. That would require either an instrument for adoption as in deHaan et al. (2024), or randomized assignment which would need a BNPL provider's cooperation.

Several confounders would remain even after these design choices. Unobserved differences in financial literacy may drive both adoption and outcomes, and the household fixed effect cannot absorb the part of financial literacy that changes over time. Differences across providers cannot be fully handled with PSID or SHED data because neither identifies the provider used. Macroeconomic shocks tied to adoption timing are absorbed by the year fixed effect only in their economy-wide component, leaving specific detail as a threat.

5. Conclusion

BNPL is a useful service in modern economics as it ties classic credit smoothing function to behavioral features that shape how people actually use it. The debate is whether BNPL smooths spending for constrained consumers and improves their welfare or whether it piles up overdraft fees and leads to a debt spiral. This proposal argues for a study of differences across households rather than average effect study. A long-panel event study with household fixed effects, explicit pre-trend tests, and an interaction by baseline liquidity would separate temporary smoothing from long-term debt accumulation. The most informative coefficients are not the average post-adoption effects, but the heterogeneity terms that show whether different households follow different paths.

The design rests primarily on the Permanent Income Hypothesis, which credit access smooths consumption and raises welfare. Under this theory, a household borrows to bridge a temporary income shortfall without changing its consumption pattern. The liquidity constraints

view of Kaplan, Violante, and Weidner (2014) built on their hand-to-mouth typology predicts that BNPL helps the wealthy hand-to-mouth but is risky for the poor hand-to-mouth. Behavioral theories of present bias (Laibson, 1997) and mental accounting (Thaler, 1985) add a further prediction that BNPL lifts spending beyond what smoothing alone would justify. The expected result is that these predictions are each partly right, but for different households. A positive heterogeneity term would show that BNPL smooths spending for asset-holding households while deepening distress for the most fragile ones.

We acknowledge several limitations, especially those from using public data. BNPL adoption is hard to measure over a long horizon and proprietary banking data would allow a stronger implementation. A partnership with a provider with real BNPL records would give a more complete picture of the product's effect. Even then, financial literacy, product differences, and macroeconomic shocks could still bias the estimates. Within these limits, the proposal identifies an unresolved tension in the literature and addresses short time horizons and differences across users. The aim is not to declare BNPL universally good or bad, but to learn which households it smooths and which it pushes into deeper distress. That evidence is what would let policymakers design better rules and help households make more informed decisions about whether BNPL fits their own circumstances.

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